SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	T. Bhanu Pavan Varma	**Reg. No.:**	192425380
Section / Batch:	B. Tech AI&ML (2024-2028)	**Date:**	14/08/2026

Code of Conduct

I T. Bhanu Pavan Varma (192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.
4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Q1. Semantic Representation in Customer Support Chatbots

The given chatbot representations are:

- ACTIVATE(Roaming, Customer)
- DEACTIVATE(CallerTune, Customer)
- QUERY(DataBalance, Customer)
- ACTIVATE(5GService, Customer)

The actual and predicted intents show that **Q2 is incorrectly interpreted**: the actual intent is *Deactivate Caller Tune*, while the predicted intent is *Activate Caller Tune*.

Justification

The semantic representation separates a customer request into an **action** and an **object**. For example:

Activate international roaming
↓
ACTIVATE(Roaming, Customer)

This representation is useful because the chatbot can directly map the user's request to an operation. In Q2, the object CallerTune is correctly identified, but the action changes from DEACTIVATE to ACTIVATE. Therefore, the system understands **what service** the customer is talking about but misunderstands **what the customer wants to do**.

The remaining three queries are correctly interpreted. Hence, there are **3 correct predictions out of 4**, giving:

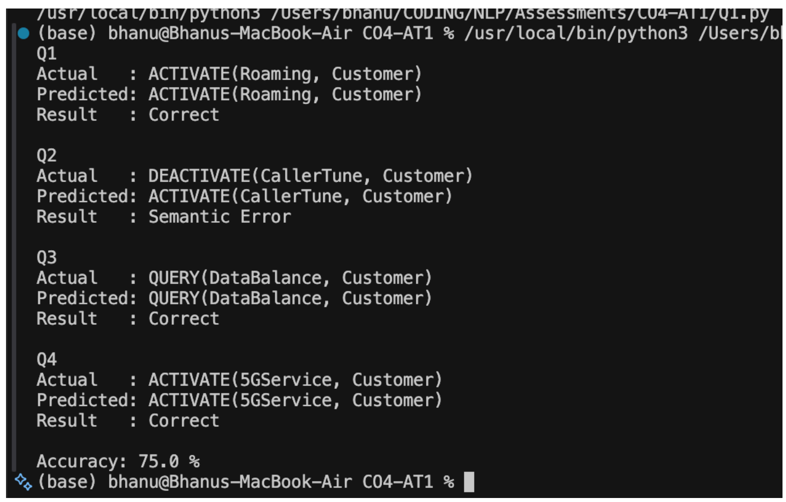
Accuracy=43 ×100=75%

Therefore, improving action/intent recognition, especially distinguishing opposite actions such as **activate vs deactivate**, would improve chatbot reliability.

Code

```
queries = {
    "Q1": {
        "actual": ("ACTIVATE", "Roaming"),
        "predicted": ("ACTIVATE", "Roaming")
    },
    "Q2": {
        "actual": ("DEACTIVATE", "CallerTune"),
        "predicted": ("ACTIVATE", "CallerTune")
    },
    "Q3": {
        "actual": ("QUERY", "DataBalance"),
```

```
        "predicted": ("QUERY", "DataBalance")
    },
    "Q4": {
        "actual": ("ACTIVATE", "5GService"),
        "predicted": ("ACTIVATE", "5GService")
    }
}

correct = 0

for q, data in queries.items():
    actual = f"{data['actual'][0]}({data['actual'][1]}, Customer)"
    predicted = f"{data['predicted'][0]}({data['predicted'][1]}, Customer)"

    print(q)
    print("Actual   :", actual)
    print("Predicted:", predicted)

    if data["actual"] == data["predicted"]:
        print("Result   : Correct")
        correct += 1
    else:
        print("Result   : Semantic Error")

    print()

accuracy = correct / len(queries) * 100

print("Accuracy:", accuracy, "%")
```



Conclusion

The semantic representation system performs correctly for **Q1, Q3, and Q4**, while **Q2 demonstrates an action-level semantic error**. Improving action recognition and contextual understanding would increase chatbot decision accuracy.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $Active(x) \rightarrow Producing(x)$
- $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$
- $Maintenance(x) \rightarrow \neg Producing(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

Q2. First-Order Predicate Calculus for Smart Manufacturing

The given machine states are:

M1 → Active
M2 → Active
M3 → Maintenance
M4 → Active

The rules are:

$Active(x) \rightarrow Producing(x)$ $Produces(x,y) \wedge Active(x) \rightarrow Available(y)$ $Maintenance(x) \rightarrow \neg Producing(x)$
These are the exact production facts and rules supplied in the assessment.

Justification

Using the first rule:

$Active(x) \rightarrow Producing(x)$
we can infer:

$Producing(M1)$ $Producing(M2)$ $Producing(M4)$
For M3:

$Maintenance(M3)$
therefore:

$\neg Producing(M3)$
So, **M1, M2, and M4 are producing**, while **M3 is not producing**.

However, the question does **not provide any Produces(machine, product) facts**. Therefore, we cannot logically determine which specific products are available or whether **Gear** is produced by M3.

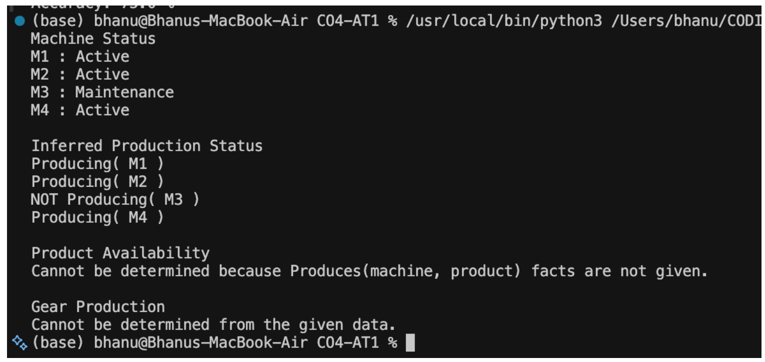
For example, if the missing fact were:

$Produces(M3,Gear)$
then maintenance of M3 would affect Gear production. But that fact is not given, so claiming that Gear production is definitely affected would go beyond the supplied data.

Code

```
machines = {
    "M1": "Active",
    "M2": "Active",
```

```
    "M3": "Maintenance",
    "M4": "Active"
}

producing = set()

for machine, status in machines.items():

    if status == "Active":
        producing.add(machine)

print("Machine Status")

for machine, status in machines.items():
    print(machine, ":", status)

print("\nInferred Production Status")

for machine in machines:
    if machine in producing:
        print("Producing(", machine, ")")
    else:
        print("NOT Producing(", machine, ")")

print("\nProduct Availability")
print("Cannot be determined because Produces(machine, product) facts are not given.")

print("\nGear Production")
print("Cannot be determined from the given data.")
```



Conclusion

Predicate calculus provides a precise framework for industrial reasoning, but its conclusions are limited by the completeness of the available production data.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Q3. Word Sense Disambiguation in E-Commerce Search Engines

The assessment provides four ambiguous queries and their clicked results.

Query	Clicked Result	Correct Sense
Apple accessories	iPhone Charger	Technology Brand
Mouse wireless	Bluetooth Mouse	Computer Device
Java tutorial	Coding Lessons	Programming Language
Python course	Software Development Training	Programming Language

Justification

Each word has more than one possible meaning. Therefore, the search engine must use **contextual clues** to select the correct sense.

For example:

Apple accessories
↓
iPhone Charger
↓
Technology Brand

The word **Apple** could mean a fruit, but *iPhone Charger* clearly indicates the technology brand.

Similarly:

Mouse wireless
↓
Bluetooth Mouse
↓
Computer Device

The query context and clicked result indicate a computer mouse rather than an animal.

For **Java tutorial** and **Python course**, words such as *tutorial*, *course*, and the clicked coding/software results indicate the programming-language senses.

Therefore, all four queries can be correctly disambiguated using **query context + clicked-result information**.

Code

```
data = {
    "Apple accessories": {
        "senses": ["Fruit", "Technology Brand"],
        "result": "iPhone Charger",
        "correct": "Technology Brand"
    },
    "Mouse wireless": {
        "senses": ["Animal", "Computer Device"],
        "result": "Bluetooth Mouse",
        "correct": "Computer Device"
    },
    "Java tutorial": {
        "senses": ["Island", "Programming Language"],
        "result": "Coding Lessons",
        "correct": "Programming Language"
    },
    "Python course": {
        "senses": ["Snake", "Programming
Language"],
        "result": "Software Development
Training",
        "correct": "Programming Language"
    }
}

for query, info in data.items():

    print("Query:", query)
    print("Possible Senses:", ",
".join(info["senses"]))
    print("Clicked Result:", info["result"])
    print("Selected Sense:", info["correct"])
    print()
```

Conclusion

The given search logs show that contextual information and clicked results successfully resolve the ambiguity of all four words. A large-scale WSD system should combine linguistic context with behavioral and domain information to improve recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1.Analyze how syntactic structures contribute to semantic role assignment.

2.Determine whether the assigned semantic roles are appropriate.

3.Evaluate potential errors that could arise from incorrect parsing.

4.Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Q4. Syntax-Driven Semantic Analysis in Healthcare

The assessment provides these parsed sentences:

1. **Doctor prescribed medicine to patient.**
2. **Patient reported severe headache.**
3. **Nurse monitored patient continuously.**
4. **Medicine reduced blood pressure.**

It also provides the semantic roles:

Doctor → Agent
Medicine → Instrument
Patient → Recipient
Headache → Symptom

Justification

Syntactic structure helps identify the relationship between entities.

For example:

Doctor → prescribed → medicine → to patient

The **Doctor** performs the action, so Doctor is the **Agent**. The **Patient** receives the medicine, so Patient is the **Recipient**. The **Headache** represents a medical symptom, so Headache is correctly classified as a **Symptom**.

The given role **Medicine → Instrument** is less straightforward. In:

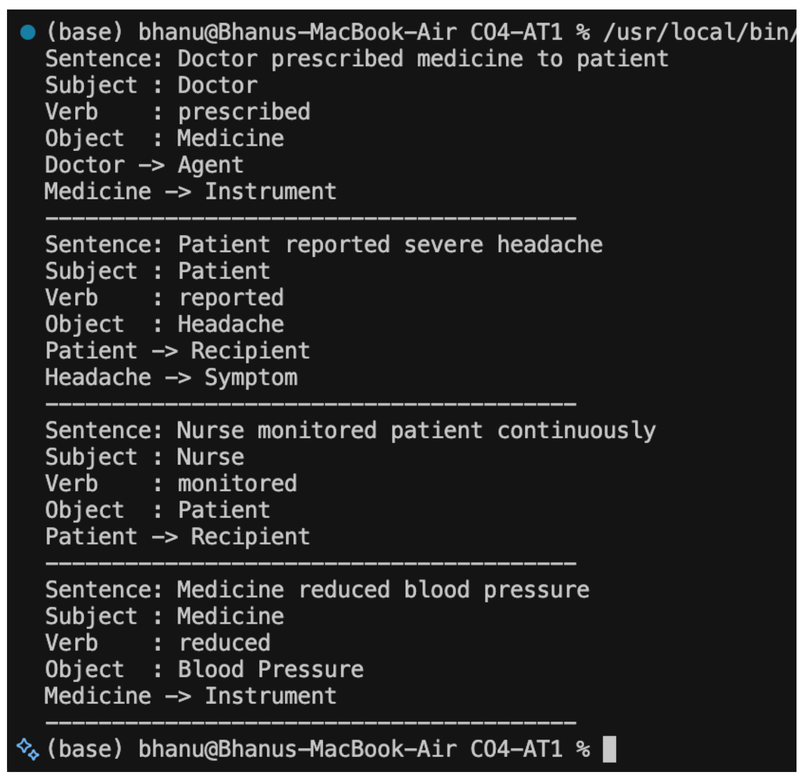
Medicine reduced blood pressure.

medicine is more naturally interpreted as a **Cause** rather than an instrument used by an agent. Therefore, this is an area where syntax alone may not provide the complete semantic interpretation.

Incorrect parsing could cause serious errors in healthcare applications, such as confusing the doctor and patient roles or incorrectly extracting medication information.

Code

```
sentences = [
    {
        "sentence": "Doctor prescribed medicine to patient",
        "subject": "Doctor",
        "verb": "prescribed",
        "object": "Medicine"
    },
    {
        "sentence": "Patient reported severe headache",
        "subject": "Patient",
        "verb": "reported",
        "object": "Headache"
    },
    {
        "sentence": "Nurse monitored patient continuously",
        "subject": "Nurse",
        "verb": "monitored",
        "object": "Patient"
    },
    {
        "sentence": "Medicine reduced blood pressure",
        "subject": "Medicine",
        "verb": "reduced",
        "object": "Blood Pressure"
    }
]

roles = {
    "Doctor": "Agent",
    "Medicine": "Instrument",
    "Patient": "Recipient",
    "Headache": "Symptom"
}

for item in sentences:

    print("Sentence:",
item["sentence"])
    print("Subject :", item["subject"])
    print("Verb    :", item["verb"])
    print("Object  :", item["object"])

    subject = item["subject"]
    obj = item["object"]
```



```
(base) bhanu@Bhanus-MacBook-Air C04-AT1 % /usr/local/bin/
Sentence: Doctor prescribed medicine to patient
Subject : Doctor
Verb    : prescribed
Object  : Medicine
Doctor -> Agent
Medicine -> Instrument
----------------------------------------
Sentence: Patient reported severe headache
Subject : Patient
Verb    : reported
Object  : Headache
Patient -> Recipient
Headache -> Symptom
----------------------------------------
Sentence: Nurse monitored patient continuously
Subject : Nurse
Verb    : monitored
Object  : Patient
Patient -> Recipient
----------------------------------------
Sentence: Medicine reduced blood pressure
Subject : Medicine
Verb    : reduced
Object  : Blood Pressure
Medicine -> Instrument
----------------------------------------
(base) bhanu@Bhanus-MacBook-Air C04-AT1 %
```

```
    if subject in roles:
        print(subject, "->", roles[subject])

    if obj in roles:
        print(obj, "->", roles[obj])

    print("-" * 40)
```

Conclusion

Syntax-driven semantic analysis provides the structural foundation for understanding medical sentences. However, syntax alone may not always distinguish roles such as **Instrument and Cause**, so combining dependency parsing, semantic-role labeling, medical knowledge, and contextual models can improve accuracy and reliability.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________